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Attitudes towards feminism predict susceptibility to feminism-related fake news

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Summary

False memories may be especially likely when one is exposed to misinformation that is consistent with one's beliefs. Here, we assessed whether feminist attitudes predict susceptibility to feminism-related fake news. In Experiment 1 ($n = 1537$), the more negative participants' attitudes towards feminism, the more likely they were to report a false memory for a fabricated event that negatively reflected on the feminist movement, and vice-versa. This effect was only evident for those who interpreted the event as expected (e.g., those who rated the event as bad for feminism). When the purpose of the study was revealed, feminist attitudes also predicted ability to identify the stories as fake. We replicated these findings in Experiment 2 ($n = 786$), using fake stories that were less ambiguous. This study suggests that individuals are more susceptible to false memories for fake news stories that are ideologically congruent, even after a warning.

KEYWORDS

fake news, false memory, memory, politics

1 | INTRODUCTION

There is some debate as to whether social media encourages us to live our online lives in “ideological echo chambers” or “bubbles” and distorts our perceptions of the world (Eady et al., 2019). However, if we did all see the same news, would we be likely to form different memories of the past? Decades of research have demonstrated the ease with which individuals can form rich false memories and beliefs for events that never happened (Loftus, 2005), often in response to fake news stories (Greene & Murphy, 2020) and that these memories can be biased in line with our political opinions (Murphy et al., 2019). The current study assessed false memories and beliefs for feminism-related events that never occurred, examining whether susceptibility to these fake news stories differs according to existing attitudes towards feminism.

The biasing effect of political opinions can be understood within a number of memory models. The Source Monitoring Framework

(Johnson et al., 1993) explains that our memories are not stored with individual files with tags that identify their source, we must infer the source of information based on the available information. So, for example, when establishing whether we witnessed a conversation between two people or we heard about it second-hand, we engage in two forms of judgement; heuristic (the amount of perceptual information available, or how closely it matches with a schema or template – “Does that sound like X's voice?”) and systematic (retrieval of supporting memories, reasoning about inconsistencies between the memory and what is otherwise known – “Does this conversation fit with what I know about the relationship I have with X?”). Source monitoring is generally very effective and we are typically successful at distinguishing the source of information. However, false memories are likely to occur when these source monitoring judgements are distorted, for example when the fabricated event appears more perceptually detailed through the use of doctored photos or encouraged imagination (Henkel, 2011; Sacchi et al., 2007; Thomas et al., 2003;

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Wade et al., 2002) or when the fabricated event is in line with our current opinions or beliefs (Montgomery & Rajagopal, 2018). Political bias then could be one manner in which source monitoring errors may be increased, as political opinions may weight expectations in a particular direction. Evidence suggests that when an individual has insufficient information to determine the source of a memory, they are likely to guess based on prior knowledge (Bayen et al., 2000), using stereotypes and schemata to determine the most likely source (Spaniol & Bayen, 2002). This results in a greater likelihood of stereotype-consistent false memories (Kleider, Goldinger, & Knuysky, 2008; Kleider, Pezdek, et al., 2008). Individuals may also engage in “content borrowing”, where experiential details are imported from true memories and can increase confidence that the false memory is a true recollection (Lampinen et al., 2005; Lyle & Johnson, 2006). Thus there are a number of ways that political bias may affect how individuals reconstruct events, as well as affecting their confidence in these memories.

A number of applied studies have found evidence to support this biasing effect in the context of politics, demonstrating the importance of the match between one's own beliefs and a fabricated news story. Conservatives were found to be more likely to report a false memory for a fabricated scandal involving President Obama than were liberals, and liberals were more likely to report a false memory for a fabricated scandal involving President Bush than were conservatives (Frenda et al., 2013). This “congruency effect” - where alignment of prior beliefs with a fake story increases false memory susceptibility - was recently demonstrated during a real-world abortion referendum (Murphy et al., 2019). Yes and No voters were presented with fabricated news stories 1 week before the referendum and the findings indicated that Yes voters were more susceptible to false memories and beliefs for fabricated No scandals, and No voters were more susceptible to false memories and beliefs for fabricated Yes scandals. When informed about the purpose of the study, Yes voters were less likely to correctly identify fabricated No scandals as fake, and No voters were less likely to correctly identify fabricated Yes scandals as fake. These studies suggest that existing attitudes can affect susceptibility to false memories and impact the ability to detect fake news even when alerted to it.

Though many political decisions are necessarily binary (Clinton vs. Trump, Yes vs. No, Leave vs. Remain, etc.), political opinions are often more complex. For instance, a voter could be slightly leaning towards a candidate, or an ardent supporter who is deeply involved in the campaign. To date, the studies that have demonstrated an effect of ideological congruency on false memories have only examined this effect along a binary (Yes vs. No voters, Liberals vs. Conservatives) and so it is not clear if this effect would be moderated by strength of support. Though there is evidence that Yes supporters would be more susceptible than No supporters to false memories or beliefs for a No campaign scandal, it is not clear whether a deeply-committed Yes supporter would be more susceptible than a wavering Yes supporter. Such a continuum would make sense, as those who feel stronger about an issue may experience a greater biasing effect when making a judgement about the source of the memory.

A further issue is the interpretation of news stories. Political events can be complicated and interpretation can vary between

individuals (Claassen & Ensley, 2016). In previous work (Murphy et al., 2019), qualitative data indicated that though the scandals were mostly perceived as intended (i.e., as negative for the side mentioned), there were some participants who reported a neutral or apathetic response to the story, and even a minority who said that the story actually reflected well on the side mentioned. It is not clear whether interpretation of the story as intended is essential to observe the congruency effect. This is important for understanding the mechanisms behind the congruency effect. For example, are conservatives more likely to remember a fabricated scandal involving President Obama because it aligns with their views (i.e., it reflects negatively on a Democratic president), or because it is familiar to them in some other way (i.e., similar to other stories they may have heard before, or more familiar due to differences in media consumption, etc.)? If subjective interpretation plays a role in susceptibility to false memories and beliefs, this may have implications for the design and targeting of interventions to combat misinformation. Individuals may be most susceptible to stories they interpret to reflect well on their preferred political side or reflect poorly on the opposition, as distinct from whether a third-party might view the story as objectively positive or negative for either side.

A contemporary political issue where opinions and perceptions of news stories can differ widely is feminism (Lanius, 2019; PettyJohn et al., 2019). We hypothesised that opinions about feminism would predict susceptibility to fabricated stories about feminism, with those who support feminism being more susceptible to stories that reflect well on feminism and those who do not support feminism being more susceptible to stories that reflect poorly on feminism.

2 | EXPERIMENT 1

This study was conducted in early 2019, in the wake of the #MeToo movement, which was a form of “hashtag activism” intended to demonstrate how widespread sexual abuse is in the lives of women and girls (Kangere et al., 2017). We examined three research questions:

1. Do attitudes towards feminism differentially predict false memories and beliefs for stories that reflect positively/negatively on feminism?
2. Do attitudes towards feminism predict ability to identify fake feminism-related news stories as fabricated?
3. Are these effects evident only for those who interpret the stories as expected?

2.1 | Method

2.1.1 | Participants

We recruited for this study via student email lists and social media posts and targeted as many responses as possible before an agreed stopping date. The study was completed by 1537 participants with a mean age of 26.05 years (SD = 9.22). There were no exclusion

criteria, beyond being over 18 years old. Participants were mostly (60%) women ($n = 920$), with 565 men, 25 who reported their gender as other, and 27 who preferred not to say. Most participants (86%) were Irish nationals, and the remaining participants were from another EU country (9%) or outside of the EU (5%). When asked if they identified as a feminist, 58% agreed or strongly agreed, 21% disagreed or strongly disagreed, and 21% neither agreed nor disagreed. 87% of the sample said they used social media a few times a day or more, over the past year. Participants reported interest in the #MeToo movement, with 75% saying they were somewhat likely, likely or very likely to read an online article about the #MeToo movement if they came across it.

2.1.2 | Materials

Feminism Attitudes. A scale to measure support for the feminist movement was developed for this study. Participants were asked to rate their agreement with five statements on a scale of (1) Strongly Agree to (5) Strongly Disagree; I identify as a feminist, I believe the feminist movement has gone too far (reverse scored), I support the feminist movement, I believe the feminist movement is necessary, I support the #MeToo movement. These were summed to create a Feminism Attitudes scale. Cronbach's alpha indicated high reliability ($\alpha = .91$). We purposefully did not define feminism for participants before they undertook the study, as we were interested in their existing perceptions of feminism (as they perceived it) and did not wish to influence responses.

News Stories. Participants were presented with 8 news stories in a random order. All stories were presented as an image followed by 2–3 lines of text.

True Stories. All participants saw the same six true stories; Bill Cosby convicted of sexual assault, Controversial comments about rape victims by Irish radio host George Hook, “House of Cards” cancelled after Kevin Spacey allegations emerge, Controversial comments made by actor Liam Neeson about the #MeToo movement, Google employee James Damore fired over an internal memo about discrimination, Donald Trump comments about a war on men in America. All true stories were presented with a general image of the subject of the story (e.g., a photograph of Liam Neeson giving an interview).

Fake Stories. All participants saw two fake stories; one concerning estimates of fabricated rape claims and one concerning a riot that broke out at a protest. As can be seen in Table 1, two versions of each story were created – one that aligned with feminist views (e.g., that there are low levels of fabricated rape claims, that men's rights protestors were in the wrong) and one that did not align with feminist views (e.g., that there are higher levels of fabricated rape claims, that feminist protestors were in the wrong). All participants saw the feminism-aligned version of one story and the feminism-misaligned version of the other story. Both stories were entirely false. There have never been any studies to estimate rates of fabricated rape claims in Ireland. There was an “I Believe Her” protest in Dublin, as described in the riot story, but there was no violence, no injuries and no damage to property. Both versions of the riot story were accompanied by the same image of a man and woman shouting at each other at a protest and both versions of the rape statistics story were accompanied by the same image of an Irish courthouse.

For each story, participants were asked if they remembered the event and could choose from I remember seeing/hearing about this, I do not remember seeing/hearing this but I remember it happening, I do not remember this event but I believe it happened, I remember this event differently, I do not remember this event. As in previous work (Murphy et al., 2019), participants were classed as remembering an event if they indicated that they specifically remembered seeing it or they generally remembered it happening. They were classed as not remembering an event if they said they remembered the event differently or not at all. Unless otherwise stated, participants who stated that they merely believed an event had happened were excluded from the analysis (as in Murphy et al., 2019). This is to allow an assessment of false memories for an event, separate from belief in that event, given that these are distinct processes which do not always mutually occur (Otgaar et al., 2014; Wade et al., 2018).

After each story, participants were asked “How ultimately good/bad do you believe this event was for the feminist movement?” and could answer on a slider from 0 (very bad) to 100 (very good).

After viewing all eight stories, participants were told “Some people who completed this survey were shown fabricated news stories (news stories that didn't happen, they were completely fabricated by

TABLE 1 The fake stories used in Experiment 1

	Rape statistics story	Riot story
Feminism-Aligned	<i>In October 2018, an article in the Irish Independent caused controversy by estimating that an average of just 4% of rape cases brought to the Irish courts are false.</i>	<i>Following the acquittal of charges against two Ulster Rugby Players in the Belfast rape trial in March 2018, a large-scale “I Believe Her” protest on O'Connell Street in Dublin turned violent, with Men's Rights counter-protesters assaulting a female protester, leading to a larger altercation causing extensive damage to local businesses and injuring two Gardaí.</i>
Feminism-Misaligned	<i>In October 2018, an article in the Irish Independent caused controversy by estimating that an average of 16% of rape cases brought to the Irish courts are false.</i>	<i>Following the acquittal of charges against two Ulster Rugby Players in the Belfast rape trial in March 2018, a large-scale “I Believe Her” protest on O'Connell Street in Dublin turned violent, with protestors assaulting a passer-by, setting off a riot which caused extensive damage to local businesses and injured two Gardaí.</i>

Note: All participants saw one version of the rape statistics story and the other version of the riot story. Note that “Gardaí” (mentioned in the riot stories) is the term for Irish police officers.

the researchers). If you think any of the stories you saw were fake, please select them below” and were shown all stories again.

2.1.3 | Procedure

The study was conducted entirely online, using the Qualtrics survey platform. Participants first completed the demographics and feminism attitudes questions. Then each participant viewed eight news stories (six true, two fake) presented in random order. Participants were then asked to pick out any stories that they believed were fabricated before they were debriefed. We used a debriefing procedure shown to be effective in a similar fake news study (Murphy et al., 2020). The study received ethical approval from the School of Applied Psychology Ethics Committee, University College Cork.

2.2 | Results

Over half the sample (53%) falsely claimed to remember at least one fabricated event; 39% remembered one false event and a further 14% remembered two false events. Both versions of the riot story were remembered at a high rate (Feminist Riot = 51%; Men's Rights Riot = 55%), with lower rates for both versions of the false rape claim statistics story (Low Rates = 29%; High Rates = 26%). This compares to a rate of reported memories ranging between 34% and 92% for the true stories (average = 3.6 true stories recalled, $SD = 1.49$).

When asked how good each fabricated event was for the feminist movement, the feminist riot was rated as significantly worse ($M = 36.03$, $SD = 27.60$) than the men's rights riot ($M = 52.44$, $SD = 28.91$), $t(1318) = 10.55$, $p < .001$, $d = 0.58$. The article describing high rates of false rape claims was also rated as significantly worse for the feminist movement ($M = 34.39$, $SD = 27.06$) than the article describing lower rates ($M = 45.85$, $SD = 28.80$), $t(1288) = 11.46$,

$p < .001$, $d = 0.41$. There was a very small, non-significant correlation between feminism attitudes and ratings for feminism-aligned stories ($r[1300] = -.08$, $p = .006$), with those who reported more negative attitudes towards feminism rating the stories as less positive. There was no such correlation for the feminism-misaligned events ($r[1273] = -.004$, $p = .887$). We classified participants as seeing a story as positive for feminists if it was 51 or above on the 1–100 scale and negative if it was 50 or below.¹ Collapsing both feminism-aligned stories, 55% rated the story as expected (i.e., rated either the men's rights riot or the low rates of false rape claims as positive). Collapsing both feminism-misaligned stories, 78% rated the story as expected (i.e., rated the feminist riot or the higher rates of false rape claims as negative). We split the file to separately analyse those who interpreted the events as expected (the feminism-aligned events positively/ the feminism-misaligned events negatively), vs. those who did not.

2.2.1 | Do attitudes towards feminism predict reported memories for fabricated feminism-related events?

Responses to the two feminist-aligned stories were collapsed and binary logistic regressions were conducted to assess the effect of feminist attitudes on false memories. Analyses were conducted separately for those who interpreted the feminism-aligned story as positive for the feminist movement ($n = 438$) and those who viewed the story as negative ($n = 594$). A further 184 participants declined to rate the story and so are not included in either analysis. Those who reported a mere belief in the fabricated event are also excluded here.

For those who interpreted the story as expected, the model was statistically significant, $\chi^2(1, N = 438) = 16.49$, $p < .001$, R^2 (Cox & Snell) = .04, R^2 (Nagelkerke) = .05, and correctly classified 59% of cases. As shown in Table 2, there was a significant effect of

TABLE 2 Results of four binary logistic regressions for false memories of the feminism-aligned events and separately, for the feminism-misaligned events from Experiment 1

		Predictors	<i>b</i>	<i>SE b</i>	Wald	<i>df</i>	<i>p</i>	<i>Exp (b)</i>	95% C.I. (<i>b</i>)
Feminism-Aligned Stories	Interpreted event as positive for feminism ($n = 438$)	Feminism Attitudes	−0.08	0.02	15.68	1	<.001	0.92	[0.89, 0.96]
		Constant	0.95	0.25	14.29	1	<.001	2.59	-
	Interpreted event as negative for feminism ($n = 594$)	Feminism Attitudes	0.02	0.02	1.60	1	.206	1.02	[0.99, 1.06]
Feminism-Misaligned Stories	Interpreted event as negative for feminism ($n = 817$)	Constant	−0.64	0.22	8.35	1	.004	0.52	-
		Feminism Attitudes	0.05	0.02	11.78	1	.001	1.05	[1.02, 1.08]
	Interpreted event as positive for feminism ($n = 233$)	Constant	−1.10	0.19	33.43	1	<.001	0.33	-
		Feminism Attitudes	0.00	0.03	0.02	1	.886	1.00	[0.95, 1.06]
		Constant	0.14	0.35	0.15	1	.698	1.14	-

Note: The models for those who interpreted the stories as expected are shown separately to those who did not interpret the stories as expected. Significant predictors are shown in bold.

feminism attitudes, such that for every one-point increase (i.e., more negative attitudes), the odds of reporting a memory for the feminism-aligned story were 8% lower. For those who did not interpret the story as expected, the model was not statistically significant, $\chi^2(1, N = 594) = 1.60, p = .206, R^2$ (Cox & Snell) = .00, R^2 (Nagelkerke) = .00. As shown in Table 2, there was no significant effect of feminism attitudes on memories for this group.

Identical analyses were conducted on the rate of falsely reported memories for the feminism-misaligned stories. Analyses were again conducted separately for those who interpreted the feminism-misaligned story as negative for the feminist movement ($n = 817$) and those who viewed the story as positive ($n = 233$). A further 199 participants declined to rate the story and so are not included in either analysis. Those who reported a mere belief in the fabricated event were also excluded.

For those who interpreted the story as expected, the model was statistically significant, $\chi^2(1, N = 817) = 11.88, p = .001, R^2$ (Cox & Snell) = .01, R^2 (Nagelkerke) = .02, and correctly classified 62% of cases. As shown in Table 3, there was a significant effect of feminism attitudes, such that for every one-point increase (i.e., more negative attitudes), the odds of falsely reporting a memory for the feminism-misaligned story were 5% greater. For those who did not interpret the story as expected, the model was not statistically significant, $\chi^2(1, N = 233) = 0.02, p = .886, R^2$ (Cox & Snell) = .00, R^2 (Nagelkerke) = .00. As shown in Table 2, there was no significant effect of feminism attitudes on memories for this group.

Note that participants who reported a memory of hearing about the event are classed as “remembering” in the analyses shown in Table 2 (with mere beliefs excluded). If we used a more liberal classification of memory and included those who reported merely believing the event had occurred, the false memory rate would increase to 70%

overall, with 35% reporting one false memory or belief and 35% reporting two false memories or beliefs. The regression results do not change when those who reported a memory or belief are included. For those who interpreted the story as expected, more negative feminist attitudes predicted significantly greater likelihood of a false memory or belief for the feminism-misaligned story (OR = 1.04, [95% CI: 1.01–1.07]) and reduced likelihood of a false memory or belief for the feminism-aligned story (OR = 0.91, [95% CI: 0.88–0.94]). This was not the case for those who did not interpret the event as expected.

For illustrative purposes, we grouped participants according to their score on the feminism attitudes scale (minimum 5, maximum 25); Support Feminism (5–12, $n = 917$), Neutral (13–17, $n = 379$), Oppose Feminism (18–25, $n = 216$). Figure 1 shows rates of reported memories for each fabricated event across these groups. Note that only those who interpreted the event as expected (e.g., positive events as positive for feminism) are included here, those who did not interpret the event as expected were excluded.

2.2.2 | Do attitudes towards feminism predict ability to identify fabricated feminism-related events?

After responding to all the news stories, participants were told they may have been exposed to fake news and were asked to select any stories they thought were fabricated. The fabricated stories were selected as fake by the majority of participants; feminist riot 47%, men's rights riot 47%, high false rape claims 52%, low false rape claims 58%. The true stories were selected as fake at a lower rate overall; George Hook 16%, Bill Cosby 4%, Donald Trump 12%, Google 43%, Liam Neeson 22%, Kevin Spacey 7%. Two binary logistic regressions were conducted to assess the impact of feminism attitudes on ability

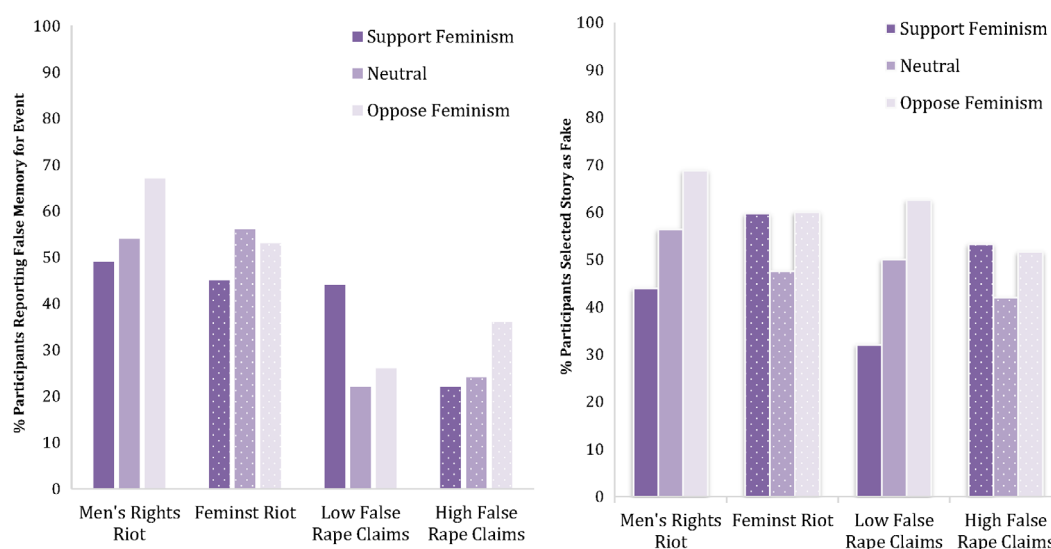


FIGURE 1 Reported memories for each of the four fabricated events (left) and correct identification of each of the four fabricated events as fake (right) in Experiment 1. Participants are grouped according to their scores on the Feminist Attitudes scale. The feminism-misaligned stories are shown in a dotted pattern

TABLE 3 Results of four binary logistic regressions for ability to correctly identify the feminism-aligned and feminism misaligned events as fabricated in Experiment 1

		Predictors	<i>b</i>	<i>SE b</i>	Wald	<i>df</i>	<i>p</i>	<i>Exp (b)</i>	95% C.I. (<i>b</i>)
Feminism-Aligned Stories	Interpreted event as positive for feminism (<i>n</i> = 577)	Feminism Attitudes	0.09	0.02	26.58	1	<.001	1.20	[1.06, 1.14]
		Constant	−1.31	0.22	34.32	1	<.001	0.27	-
	Interpreted event as negative for feminism (<i>n</i> = 724)	Feminism Attitudes	−0.00	0.02	0.07	1	.787	1.00	[0.97, 1.03]
		Constant	0.09	0.20	0.20	1	.655	1.09	-
Feminism-Misaligned Stories	Interpreted event as negative for feminism (<i>n</i> = 998)	Feminism Attitudes	−0.03	0.01	5.51	1	.019	0.97	[0.94, 1.00]
		Constant	0.50	0.17	9.09	1	.003	1.64	-
	Interpreted event as positive for feminism (<i>n</i> = 276)	Feminism Attitudes	−0.03	0.03	1.35	1	.245	0.97	[0.91, 1.02]
		Constant	−0.19	0.33	0.33	1	.567	0.83	-

Note: The models for those who interpreted the stories as expected are shown separately to those who did not interpret the stories as expected. Significant predictors are shown in bold.

to identify the story as fake. As before, these were conducted separately for those who interpreted the story as expected and those who did not.

For the feminism-aligned events, the model for those who interpreted the story as expected was statistically significant, $\chi^2(1, N = 577) = 28.22, p = <.001, R^2$ (Cox & Snell) = .05, R^2 (Nagelkerke) = .06, and correctly classified 61% of cases. As shown in Table 3, there was a significant effect of feminism attitudes, such that for every one-point increase (i.e., more negative attitudes), the odds of participants identifying the feminism-aligned story as fake were 9% greater. For those who did not interpret the story as expected, the model was not statistically significant, $\chi^2(1, N = 724) = 0.07, p = .787, R^2$ (Cox & Snell) = .00, R^2 (Nagelkerke) = .00.

For the feminism-misaligned events, the model for those who interpreted the story as expected was statistically significant, $\chi^2(1, N = 998) = 5.54, p = .019, R^2$ (Cox & Snell) = .01, R^2 (Nagelkerke) = .01, and correctly classified 52% of cases. As shown in Table 3, there was a significant effect of feminism attitudes, such that for every one-point increase (i.e., more negative attitudes), the odds of participants identifying the feminism-misaligned story as fake were 3% lower. For those who did not interpret the story as expected, the model was not statistically significant, $\chi^2(1, N = 276) = 1.38, p = .241, R^2$ (Cox & Snell) = .01, R^2 (Nagelkerke) = .01.

Participants were again grouped according to their score on the feminism attitudes scale, for illustrative purposes. Figure 1 shows the rates of correct identification for each fabricated event across these groups. Note that only those who interpreted the event as expected (e.g., positive events as positive for feminism) are included here, those who did not interpret the event as expected were excluded.

3 | EXPERIMENT 2

As expected, we found that attitudes towards feminism predicted rates of falsely reporting memories for fake news related to feminism.

Individuals with strongly feminist views were more likely to report a memory or belief for a fabricated event that reflected well on feminism and likewise, those with negative views about feminism were more likely to report a memory or belief for a fabricated event that reflected poorly on feminism. However, there were three key limitations in Experiment 1 that limit our ability to draw conclusions. Firstly, the fake stories were perhaps slightly ambiguous and a significant minority of participants did not agree with our classification of stories as reflecting well or poorly on feminism. Secondly, as all of the stories used pertained to feminism, we could not rule out the possibility that participants may have differed in their general susceptibility to false memories and beliefs, rather than feminism-specific false memories and beliefs. Finally, the explicit focus on feminism may have biased our participants. Smith et al. (2006) found that framing a survey as investigating Parkinson's disease resulted in significantly lower reported life satisfaction for Parkinson's disease patients, compared to a survey described as assessing the general population. To address these limitations, we conducted a second experiment in May 2020, investigating the same hypotheses as Experiment 1. Here, we used less ambiguous stories, compared rates of reported false memories for stories about feminists to rates of reported false memories for the same stories featuring a different group (refugees), and pitched the study as assessing memories related to the COVID-19 pandemic.

3.1 | Method

3.1.1 | Participants

The study included 802 participants who were recruited via student email lists and social media posts. Sixteen participants were excluded after they indicated they had researched on the internet or asked a friend for help while completing the survey, leaving 786 participants in the final sample, with a mean age of 33.79 ($SD = 12.12$). Participants were mostly (65%) women ($n = 512$), with $n = 264$ reporting as

men, $n = 3$ as other and $n = 7$ who preferred not to say. Most participants (91%) were Irish nationals, and the remaining participants were from another EU country (5%) or outside of the EU (4%). 90% of the sample said they used social media a few times a day or more, over the past year. When asked if they identified as a feminist, 50% agreed or strongly agreed, 22% disagreed or strongly disagreed, and 28% neither agreed nor disagreed.

3.1.2 | Materials

The survey was presented as a study of “opinions on recent news events related to the COVID-19 pandemic”.

Feminism Attitudes. Attitudes towards feminism were measured using the same scale as Experiment 1. However, in Experiment 2 the items were presented in a randomised matrix alongside other questions related to political opinions. To ensure the five feminism questions would not stand out, the filler questions also centered on specific political themes – there were five questions relating to environmentalism, six questions related to racism and xenophobia, and four questions related to trust in government and government spending. Examples of filler items are: “I think environmental protection should take precedence over economic concerns”, “All politicians are self-serving and cannot be trusted” and “I think the Black Lives Matter movement has gone too far”. Responses to these filler questions are available in our online data file. As in Experiment 1, Cronbach's alpha indicated high reliability for the feminism scale ($\alpha = .89$).

Filler questions. We also included filler questions designed to mask the purpose of the study. Participants were asked who they had trusted to advise them during the COVID-19 crisis, ranking named political parties, scientists, healthcare workers, celebrities, etc. Participants were also asked to rate whether they considered themselves to be at high risk from COVID-19. The responses to all questions are included in our online data file, but were not analysed further.

News Stories. Participants were presented with 5 news stories in a random order. All stories were presented as an image followed by 2–3 lines of text.

True Stories. All participants saw the same three true stories. The stories centered on actions taken by Irish public figures during the pandemic – inaccurate comments made by the Minister for

Health, a political party cancelling rallies due to a COVID case in the school of the party leader's children, and an athlete calling for stricter lockdowns. The images accompanying these stories was a general photograph of the subject of the story (e.g., the Minister for Health giving a speech).

Fake Stories. All participants saw two fake stories, as shown in Table 4. One was a story that depicted a group being fined for misuse of publicly-raised funds (negative story) and the other depicted a group assisting vulnerable people during the COVID-19 pandemic (positive story). Each participant saw both the positive and negative story, but were randomly assigned to either see the positive story about feminists and the negative story about refugees, or vice versa. In this way, we could compare false memory rates for identical stories, featuring different subjects. The negative story was always presented with a photograph of an elderly woman receiving a package outside her home. The positive story was always presented with a photograph of a mural tribute to front line workers.

After each story, participants were asked about how the events described reflected on the subject of the story: “How do you think this story reflects on feminists”? This was answered via a slider from 0 (very negatively) to 100 (very positively). This was a slight change in wording from Experiment 1 (where we asked participants to rate how good or bad the stories were for “the feminist movement”). The revised question more specifically assessed whether the stories reflected positively or negatively on feminists (as opposed to how they contributed to achieving feminist goals), which is more in line with previous studies of the ideological congruency effect (Murphy et al., 2019).

3.1.3 | Procedure

The study was conducted entirely online. Participants first completed the demographics and political attitudes scale before viewing the news stories, presented in random order. Participants were then asked to pick out any stories that they believed were fabricated. Before being debriefed, participants were asked “Did you use the internet or ask others to help you answer any of the questions in this survey? Your answer will not affect the rest of the survey but please answer honestly”. This was not included in Experiment 1 but was added here to improve data quality, owing to the lack of experimental

TABLE 4 The fake stories used in Experiment 2

	Subject: Feminists	Subject: Refugees
Positive Story	<i>In April, the Irish Feminists Society were praised for running a scheme to support Irish people “cocooning” at home due to COVID-19. Members delivered food and medicine to thousands of vulnerable people.</i>	<i>In April, the Irish Refugee Network were praised for running a scheme to support Irish people “cocooning” at home due to COVID-19. Members delivered food and medicine to thousands of vulnerable people.</i>
Negative Story	<i>In May, the Irish Feminist Society were fined for misuse of publicly-raised funds. The organisation ran a fundraising campaign for frontline COVID-19 workers, but an investigation found the money was used for a “wellness retreat” for members.</i>	<i>In May, the Irish Refugee Network were fined for misuse of publicly-raised funds. The organisation ran a fundraising campaign for frontline COVID-19 workers, but an investigation found the money was used for a “wellness retreat” for members.</i>

Note: All participants saw one version of the positive story and the other version of the negative story.

control in online research. The study received ethical approval from the School of Applied Psychology Ethics Committee, University College Cork.

3.2 | Results

Approximately one quarter of the sample (24%) falsely claimed to remember at least one fabricated event; 22% reported a memory for one event and a further 2% reported a memory for two events. Memories for both versions of the positive story were reported at a high rate (Feminists' COVID assistance = 20%; Refugees' COVID assistance = 32%), but memories for both versions of the negative story were reported at a lower rate (Feminists' misuse of funds = 7%; Refugees' misuse of funds = 5%). This compares to a reported memory rate of 37%, 37% and 46% for the three true stories, (average = 1.03 true stories recalled, $SD = 0.92$).

As expected, on a scale of 1–100, with 100 being very positive, the positive story was rated as reflecting well on the named group, with very similar average ratings given for the feminists version ($M = 84.09$, $SD = 18.41$) and the refugee version ($M = 83.57$, $SD = 19.70$). Likewise, the negative story was rated as reflecting poorly on the named group, though more negatively for feminists ($M = 20.47$, $SD = 19.99$) than refugees ($M = 32.16$, $SD = 21.94$). Using the same classification as Experiment 1, where a story was rated as positive for feminists if it was 51 or above on the 1–100 scale and negative if it was below 50, just 4.5% of participants rated the negative story as positive and just 3.1% rated the positive story as negative. This confirms the Experiment 2 stories were less ambiguous and more clearly positive or negative than the stories used in Experiment 1.

The feminist attitude scale was significantly correlated with ratings of how the stories reflected on feminists, for both the negative feminist story ($r(288) = -.12$, $p = .039$) and the positive feminist story, ($r(367) = -.32$, $p < .001$), with those who reported more negative attitudes towards feminism rating the stories as less positive.

3.2.1 | Do attitudes towards feminism predict reported memories for fabricated feminism-related events?

Binary logistic regressions were conducted to assess the effect of feminist attitudes on rates of falsely reporting a memory for a

fabricated event. As in Experiment 1, those who reported a mere belief in the fabricated event were excluded from this analysis.

For the positive feminist story concerning a feminist group providing assistance to vulnerable people during the COVID-19 lockdown, the model was statistically significant, $\chi^2(1, N = 277) = 4.51$, $p = .034$, R^2 (Cox & Snell) = .02, R^2 (Nagelkerke) = .03, and correctly classified 80% of cases. As shown in Table 5, there was a significant effect of feminism attitudes, such that for every one-point increase (i.e., more negative attitudes), the odds of claiming to remember the positive story were 7% lower.

For the negative feminist story, concerning the misuse of publicly-raised funds, the model was statistically significant, $\chi^2(1, N = 325) = 11.26$, $p = .001$, R^2 (Cox & Snell) = .03, R^2 (Nagelkerke) = .09, and correctly classified 93% of cases. As shown in Table 5, there was a significant effect of feminism attitudes, such that for every one-point increase (i.e., more negative attitudes), the odds of claiming to remember the negative story were 16% greater.

Crucially, the feminist attitudes scale did not significantly predict false memories for the same fake stories when the stories described actions by refugee groups, rather than feminists. We analysed this using identical binary logistic regressions. For the negative story, the model was not significant, $\chi^2(1, N = 309) = 0.07$, $p = .785$, R^2 (Cox & Snell) = .00, R^2 (Nagelkerke) = .001. For the positive story, the model was also not significant, $\chi^2(1, N = 262) = 0.62$, $p = .432$, R^2 (Cox & Snell) = .002, R^2 (Nagelkerke) = .003.

Note that participants who reported a memory of hearing about the event are classed as “remembering” in the analyses shown in Table 5 (with mere beliefs excluded). If we used a more liberal classification of memory and included those who reported merely believing the event had occurred, the reported false memory rate would increase to 51% overall, with 37% reporting one false memory or belief and 14% reporting two false memories or beliefs. As in Experiment 1, the regression results do not change when those who reported a memory or belief are included; more negative feminist attitudes predicted significantly greater likelihood of claiming a false memory or belief for the negative story ($OR = 1.07$, [95% CI: 1.01–1.13]) and reduced likelihood of a false memory or belief for the positive story ($OR = 0.93$, [95% CI: 0.88–0.97]).

For illustrative purposes, we grouped participants according to their score on the feminism attitudes scale (minimum 5, maximum 25); Support Feminism (5–12, $n = 491$), Neutral (13–17, $n = 234$), Oppose Feminism (18–25, $n = 58$). Figure 2 shows rates of reported memories for the fabricated stories across these groups.

TABLE 5 Results of two binary logistic regressions for false memories for the fabricated feminism-related news stories in Experiment 2

	Predictors	<i>b</i>	<i>SE b</i>	Wald	<i>df</i>	<i>p</i>	<i>Exp(b)</i>	95% C.I. (<i>b</i>)
Positive Feminist Story	Feminism Attitudes	−0.08	0.04	4.27	1	.039	0.93	[0.86, 1]
	Constant	−0.57	0.42	1.89	1	.173	0.57	-
Negative Feminist Story	Feminism Attitudes	0.15	0.04	11.36	1	.001	1.16	[1.06, 1.26]
	Constant	−4.42	0.65	46.56	1	<.001	0.01	-

Note: Significant predictors are shown in bold.

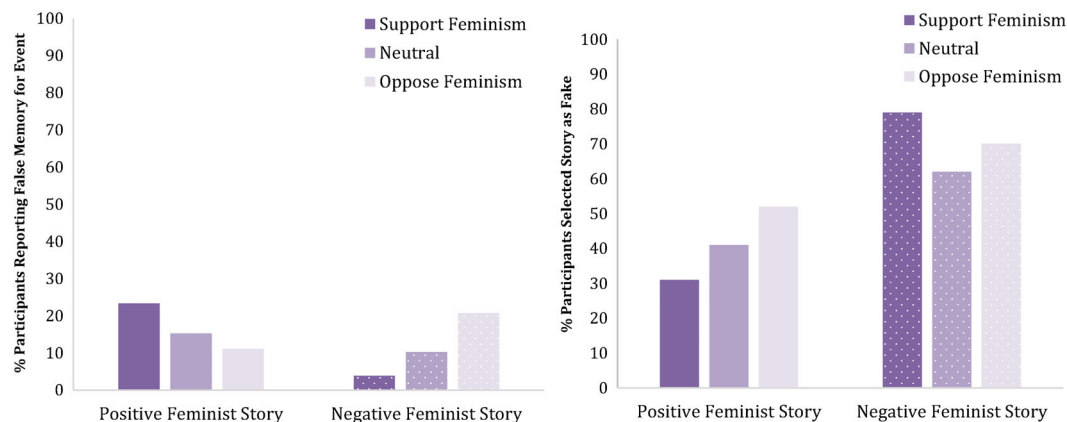


FIGURE 2 Rates of reported memories for the positive and negative feminist-related fake stories (left) and correct identification of each of the four fabricated events as fake (right) in Experiment 2. Participants are grouped according to their scores on the feminist attitudes scale, for illustrative purposes

TABLE 6 Results of two binary logistic regressions for ability to correctly identify the stories as fabricated in Experiment 2

	Predictors	<i>b</i>	<i>SE b</i>	Wald	<i>df</i>	<i>p</i>	<i>Exp(b)</i>	95% C.I.
Positive Feminist Story	Feminism Attitudes	0.06	0.03	6.02	1	.014	1.06	[1.01, 1.12]
	Constant	−1.27	0.30	18.15	1	<.001	0.28	-
Negative Feminist Story	Feminism Attitudes	−.07	0.03	7.22	1	.007	.933	[0.88, 0.98]
	Constant	1.80	0.32	31.34	1	<.001	6.10	-

Note: Significant predictors are shown in bold.

3.2.2 | Do attitudes towards feminism predict ability to identify fabricated feminism-related events?

The fabricated stories were selected as fake by the majority of participants; feminist negative story 74%, refugee negative story 70%, feminist positive story 36%, refugee positive story 29%. The true stories were selected as fake at a moderately high rate overall; athlete calls for lockdown 34%, Minister for Health error 43%, political party cancels rally 25%. For the positive story, feminism attitudes were a significant predictor of selecting the story as fake, when prompted, $\chi^2(1, N = 388) = 6.09, p = .014, R^2$ (Cox & Snell) = .02, R^2 (Nagelkerke) = .02, and correctly classified 65% of cases. As shown in Table 6, for every one-point increase (i.e., more negative attitudes), the odds of selecting the story as fake increased by 6%. Likewise, for the positive story, feminism attitudes were a significant predictor of selecting the story as fake, $\chi^2(1, N = 381) = 7.24, p = .007, R^2$ (Cox & Snell) = .02, R^2 (Nagelkerke) = .03, and correctly classified 74% of cases. As shown in Table 6, for every one-point increase (i.e., more negative attitudes), the odds of selecting the story as fake decreased by 6%.

The feminist attitudes scale was not a significant predictor of identifying the stories as fake when the stories related to refugees rather than feminists. For the negative story, the logistic regression model was not significant, $\chi^2(1, N = 388) = 1.18, p = .278, R^2$ (Cox & Snell) = .003, R^2 (Nagelkerke) = .004. For the positive story, the model was also not significant, $\chi^2(1, N = 381) = 0.77, p = .380, R^2$ (Cox & Snell) = .002, R^2 (Nagelkerke) = .003.

Participants were again grouped according to their score on the feminism attitudes scale, for illustrative purposes. Figure 2 shows the rates of correct identification for each fabricated event across these groups.

4 | EXPERIMENT 2 DISCUSSION

We replicated the findings of Experiment 1 in Experiment 2, finding that attitudes towards feminism predicted rates of reported memories for feminism-related fabricated events. Strong support for feminism was associated with an increased likelihood of reporting a memory for a fake story that reflected well on feminism, but decreased likelihood of reporting a memory for a fake story that reflected poorly on feminism. Equally, negative feelings towards feminism was associated with an increased likelihood of reporting a memory for a fake story that reflected poorly on feminism and a decreased likelihood of reporting a memory for a fake story that reflected well on feminism. This is in line with our hypotheses and prior work which has demonstrated political congruency effects in false memories and beliefs for US politics (Frenda et al., 2013) and an abortion referendum (Murphy et al., 2019). Crucially, Experiment 2 demonstrated that attitudes towards feminism did not predict rates of reported memories for identical fake stories when they were presented with refugees as the subject instead of feminists. This suggests that the results observed in this study are due to alignment with existing attitudes, rather than any other aspect of the narratives used.

5 | GENERAL DISCUSSION

As in many prior studies, the current research presents evidence that individuals can form false memories and beliefs for fabricated events. Research has identified a number of factors that may increase susceptibility towards false memories and beliefs in response to misinformation, including age, cognitive ability, subject knowledge, and analytical reasoning (Greene & Murphy, 2020; Roediger & Geraci, 2007; Zhu et al., 2010). The current study contributes to growing evidence that partisanship may also be a predictor of susceptibility to fabricated political stories (Frenda et al., 2013; Greene et al., 2021; Murphy et al., 2019). Our findings suggest that the more supportive one feels towards feminism, the more likely they are to claim to remember an event that positively reflected on feminism and the less likely they were to claim to remember an event that negatively reflected on feminism. The findings that positive and negative political opinions predict susceptibility to false memories and beliefs for related stories is in keeping with the source monitoring framework. Individuals may be more likely to suffer from source monitoring failures for attitudinally congruent information (Johnson et al., 1993), as their prior opinions may scaffold their memory and make it seem more likely to be true. Research has demonstrated that stories that are in line with one's beliefs may be supported by schemata or stereotypes (Kleider, Goldinger, & Knuycky, 2008; Spaniol & Bayen, 2002) or "borrow" from true memories (Lampinen et al., 2005). Interestingly, the biasing of source monitoring judgements by political opinions seems to be sufficiently strong that it is not overcome when stricter source monitoring is encouraged. In both of our experiments, we warned participants they may have been exposed to fake news and asked them to select any fake stories. Previous studies have shown that fake news warnings reduce belief in fake news only modestly (Clayton et al., 2019; Pennycook et al., 2020). In the current study, warnings did not eliminate the observed congruency effects. This echoes the findings of Murphy et al. (2019), suggesting that political orientation biases source judgements in a manner that may be difficult to overcome with mere warnings.

Though the current study used a simple paradigm for assessing false memories for political events that has been utilised in similar studies (Frenda et al., 2013; Greene & Murphy, 2020), we are limited in our understanding of how rich these recollections were. Future research might examine this political congruency effect in a manner that allows for deeper exploration of the phenomenology of these memories, such as interviews or using a longitudinal design to assess the memories over time. A further potential limitation is the nature of our sample, who were over 60% female in both studies. However, we would note that female and feminist are not synonymous and indeed the correlation between gender and feminist attitudes in the current study was weak (Experiment 1: $r = .30$. Experiment 2: $r = .33$). As our hypotheses relate specifically to feminist attitudes rather than sex or gender, we do not have any reasons to suspect the results would be different with a more balanced sample.

A practical recommendation for future research in this area is to consider controlling for how participants interpret the fake political

stories used to implant memories. Interpretation can be difficult to predict and is itself subject to partisan bias, for instance, voters express more concern about a scandal perpetrated by the opposing party than the same scandal committed by their own party (Claassen & Ensley, 2016). The current findings suggest that political congruency effects are only evident when participants interpret the story as expected, thus it may be an important variable to record.

The current study suggests that those with strong opinions on a given political issue may be especially likely to report false memories or beliefs for fabricated news events that align with their views. Even when alerted to the possible presence of fake news, those with stronger opinions were less likely to correctly identify the fabricated stories. The role of partisan bias should be considered when designing interventions to reduce susceptibility to misinformation.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

ENDNOTE

¹ It could be argued that 50% is a mid-point and does not reflect a response that is either positive or negative. We therefore reanalyzed our data, excluding those who gave a rating of 50. This did not significantly alter our findings, so we report our original analyses here.

DATA AVAILABILITY STATEMENT

The data and materials for this study are available at <https://osf.io/p8vmf/>.

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